

Lecture 7: Alpha Decay, Uranium Chains, and the Radon Hazard: Notes

CBE 30235: Introduction to Nuclear Engineering — D. T. Leighton

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1 The Physics of Alpha Emission (Lamarsh §2.8)

Alpha decay is the emission of a ${}^4_2\text{He}$ nucleus (2 protons, 2 neutrons). This is the dominant decay mode for heavy nuclei ($A > 200$) because it is the most efficient way for a massive nucleus to shed excess mass and charge.

1.1 Tunneling and Monoenergeticity

Unlike beta decay, which shares energy with a neutrino, alpha decay is a two-body problem. Consequently, alpha particles are **monoenergetic**—they carry a specific kinetic energy characteristic of the parent-daughter transition.

- **Quantum Tunneling:** Classically, an alpha particle is bound by the strong nuclear force. George Gamow (1928) showed that the alpha particle "tunnels" through the Coulomb barrier.
- **Geiger-Nuttall Law:** There is a direct mathematical relationship between the decay constant (λ) and the energy of the alpha particle (E). High-energy alphas tunnel more easily, leading to much shorter half-lives.

1.2 The Isotopic Zig-Zag: Moving Toward Stability

To understand why heavy chains (like Uranium) involve both alpha and beta decays, we must look at the **Neutron-to-Proton (N/Z) ratio**.

- **Alpha Emission and the Ratio:** An alpha particle removes 2 protons and 2 neutrons. Because $N > Z$ for heavy nuclei, removing equal numbers of both actually **increases** the N/Z ratio.
- **The Valley of Stability:** For heavy atoms, the "ideal" stable ratio is approximately 1.5 (e.g., Lead-206 has $124/82 \approx 1.51$). If an alpha decay pushes the nucleus too far to the neutron-rich side of the valley, the nucleus becomes unstable to **Beta Minus (β^-)** decay.
- **Beta's Counter-Correction:** A β^- decay converts a neutron into a proton, effectively **decreasing** the N/Z ratio and shifting the atom back toward the center of the valley.

1.3 The Search for Stability

In the Uranium decay series, the atom "zig-zags" down the chart of nuclides. The alpha decays reduce the total mass, while the beta decays "fine-tune" the charge-to-mass ratio. However, for elements with $Z > 82$, there is no truly stable configuration; the nucleus is simply too large for the short-range Strong Force to overcome the long-range Coulomb repulsion of the protons. The chain only terminates when it reaches **Lead** ($Z = 82$), which sits at a "magic number" of protons and represents a stable local minimum in the potential energy surface.

1.4 Linear Energy Transfer (LET) and Range

From a safety engineering perspective, alpha particles are high-LET radiation.

- **Mass and Charge:** Being heavy and +2 charged, they interact strongly with electrons in matter.
- **The Bragg Peak:** They deposit nearly all their energy at the very end of their track. In tissue, an alpha particle typically travels only **40–80 μm** , about the width of a few human cells.
- **The Shielding Paradox:** A sheet of paper or the dead layer of skin (*stratum corneum*) is sufficient to stop an alpha particle. Thus, they are an **internal hazard** only (via inhalation or ingestion), but once inside, they are roughly 20 times more biologically damaging than gammas or betas.

References for Further Reading: Tunneling and Alpha Kinetics

- **Gamow, G. (1928):** "[The Quantum Theory of Nuclear Disintegration.](#)" *Nature*, 122, 805. This is the seminal paper that applied the new wave mechanics to the nucleus, proving that alpha particles tunnel through a potential barrier.
- **The Geiger-Nuttall Law:** [Geiger, H., and Nuttall, J. M. \(1911\).](#) "[The ranges of the \$\alpha\$ particles from uranium.](#)" *Philosophical Magazine*, 22(130), 612-621. This empirical law, discovered before quantum mechanics, showed that $\log(\lambda)$ is proportional to $\log(E)$, a mystery Gamow's tunneling theory eventually solved.
- **Physics Libre Texts:** [Alpha Decay](#). A somewhat technical discussion of the connection between Gamow's theory and the Geiger-Nuttall Law.

2 Uranium and Thorium Decay Series

The Earth's natural radioactive background is dominated by three "primordial" decay chains. These chains follow the **$4n$ rule**, where the mass number A of every isotope in a series can be described by a specific mathematical remainder when divided by 4.

2.1 The Three Extant Natural Chains

As discussed in **Lamarsh Section 9.7**, there are three main series still found in nature:

1. **The Uranium Series ($4n + 2$):** Begins with ^{238}U , passes through ^{226}Ra and ^{222}Rn , and terminates at ^{206}Pb .

2. **The Thorium Series ($4n$):** Begins with ^{232}Th and terminates at ^{208}Pb . The key daughter is **Radon-220** (Thoron), which has a half-life of only 55.6 seconds.
3. **The Actinium Series ($4n + 3$):** Begins with ^{235}U and terminates at ^{207}Pb . It produces **Radon-219** (Actinon), with a half-life of 3.96 seconds.

2.2 Secular Equilibrium and the "Radon Leak"

In geological formations, these chains exist in **Secular Equilibrium** ($A_{\text{parent}} = A_{\text{daughter}}$). This is a consequence of steady-state: the rate of production of each intermediate must equal the rate of decay. However, the equilibrium may be physically broken at the Radon step.

- **Noble Gas Exception:** Radon is the only element in these chains that does not form chemical bonds with the surrounding rock matrix.
- **Emanation:** While its "parents" (Uranium and Radium) are locked in the mineral grains, Radon can escape into the pore spaces between grains.

2.3 Concentration by Human Activity: The Coal Cycle

While these elements are "natural," engineering processes can concentrate them to hazardous levels. As noted in **USGS Fact Sheet 163-97**, burning coal for power concentrates the non-volatile Uranium and Thorium into the **fly ash** by a factor of 10 or more relative to the raw coal. This "Technologically Enhanced Naturally Occurring Radioactive Material" (TENORM) is a major focus of coal ash waste management.

References for Section 2

- Lamarsh, J. R., & Baratta, A. J.: *Introduction to Nuclear Engineering*, 4th Ed., **Section 9.7**.
- **USGS Publications Warehouse:** [FS-163-97: Radioactive Elements in Coal and Fly Ash](#).
- **Wikipedia:** [Decay Chain](#) (An excellent discussion of the key decay chains).

3 The Radon Problem: Transport and Ingress

Radon-222 (^{222}Rn , $T_{1/2} = 3.82$ days) is the most significant source of human exposure to ionizing radiation. Because it is a noble gas, it acts as a mobile link in the Uranium decay chain, escaping the solid earth to enter the human environment.

3.1 Emanation and Diffusion

The journey of Radon begins with **emanation**: the physical escape of the Radon atom from a mineral grain into the surrounding pore space.

- **Recoil Mechanism:** When ^{226}Ra decays, the resulting ^{222}Rn atom recoils with enough energy (~ 100 keV) to travel roughly 30 nm in rock. If the decay occurs near the grain surface, the atom "kicks" itself into the pore.
- **Diffusion (Fick's Law):** In the absence of pressure gradients, Radon moves through the soil pore air via molecular diffusion. The diffusion length (L) is given by $L = \sqrt{D/\lambda}$, where D is the diffusion coefficient and λ is the decay constant. For most soils, $L \approx 1$ meter.

3.2 Advection: The "Sump" Effect

While diffusion exists, the primary driver for high indoor Radon levels is **advection** (bulk fluid flow).

- **Pressure Gradients:** A house often operates at a slightly lower pressure than the surrounding soil (the "Stack Effect"). Warm air rising or wind blowing across the roof creates a vacuum in the basement (typically 1–10 Pa).
- **Darcy's Law:** The flow rate of soil gas (J) into the basement is governed by the soil permeability (κ) and the pressure gradient (∇P):

$$J = -\frac{\kappa}{\mu} \nabla P \quad (1)$$

where μ is the viscosity of the soil gas. Even tiny cracks in a concrete slab act as "high-conductivity" channels for this gas.

3.3 Concentration Units: The Curie vs. The Becquerel

In the US, Radon is measured in **picoCuries per liter (pCi/L)**.

- **EPA Action Level:** 4.0 pCi/L ($\approx 148 \text{ Bq/m}^3$).
- **Outdoor Baseline:** Typical outdoor air is $\sim 0.4 \text{ pCi/L}$.
- **Indoor Extremes:** In regions with high Uranium content (e.g., the Reading Prong in PA), levels can exceed 1,000 pCi/L, equivalent to smoking hundreds of cigarettes a day.

3.4 The Watras Case: From Miners to Homeowners

Historically, Radon was viewed as a localized risk for underground miners. This changed in December 1984 due to the experience of Stanley Watras, an engineer at the Limerick Nuclear Power Plant (Pennsylvania).

- **The Detection Paradox:** Watras triggered the plant's radiation alarms while *entering* the facility. Because the plant's monitors were highly sensitive to alpha-emitting progeny on his clothing, they detected the "background" he was bringing from home.
- **The Magnitude:** His home was found to have a concentration of 2,700 pCi/L. This remains one of the highest indoor radon measurements ever recorded in a residential structure.
- **Regulatory Impact:** This event led directly to the **Indoor Radon Abatement Act (IRAA)**, establishing the national goal that indoor air be as free of radon as the ambient outdoor air.

3.5 Radon Progeny and Lung Cancer

A common misconception is that Radon gas itself causes cancer. Because it is a noble gas with a relatively long half-life, most inhaled ^{222}Rn is simply exhaled. The dose is delivered by the **Radon Progeny** (Short-lived daughters).

- **The Daughters:** ^{222}Rn decays into solid, chemically reactive isotopes: ^{218}Po , ^{214}Pb , ^{214}Bi , and ^{214}Po .

- **The Working Level (WL):** In engineering practice, we use the "Working Level" to describe the concentration of short-lived progeny. Because of ventilation and "plate-out" (daughters sticking to walls), the daughters are rarely in secular equilibrium with the Radon gas.
- **Mechanism of Injury:** These solid particles attach to aerosols (dust/smoke). When inhaled, they "plate out" on the bronchial epithelium. The alpha decays of ^{218}Po and ^{214}Po deliver a high-LET dose directly to the nuclei of the basal cells.

3.6 Engineering Mitigation

Reducing Radon exposure is a matter of pressure management and ventilation.

- **Sub-Slab Depressurization:** A fan is used to create a vacuum under the basement floor, intercepting Radon gas before it enters the structure and venting it above the roofline.
- **Sealing:** Closing "preferential pathways" like French drains, sump pits, and floor cracks.

References for Section 3

- **Nazaroff, W. W. (1992):** "[Radon Transport from Soil to Air.](#)" *Reviews of Geophysics*, 17, 1-18. (The definitive engineering review on transport math).
- **EPA Assessment:** "[Health Risk of Radon.](#)" Provides the statistical basis for the 4.0 pCi/L action level.
- **Lamarsh, J. R., & Baratta, A. J.:** *Introduction to Nuclear Engineering*, 4th Ed., **Section 9.7.**
- **radon-ohio.com:** "[The Stanley Watras Story.](#)" (Account of the discovery of Radon contamination in the home).

4 Conclusion: The Geology of Health

The Radon hazard illustrates that nuclear engineering is not confined to reactors; it is a fundamental part of our built environment. Understanding the decay kinetics of the Uranium series allows us to quantify and mitigate a major public health risk.